

An Orlicz-space characterization for the class $\mathcal{X}$

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Source and target

The source paper is Astashkin–Lykov–Milman, *Majorization revisited: Comparison of norms in interpolation scales*, arXiv:2107.11854 [1]. In Section 6 the authors define the class $\mathcal{X}$ of rearrangement invariant spaces X on $(0, \infty)$ with the following property. If a pair (f, g) satisfies the Nazarov–Podkorytov single-crossing condition

$$f^*(t) \geq g^*(t) \quad (t \leq t_0), \quad f^*(t) \leq g^*(t) \quad (t \geq t_0),$$

and if $\|f\|_X \geq \|g\|_X$, then

$$\|f^* \chi_{(0,s)}\|_X \geq \|g^* \chi_{(0,s)}\|_X \quad \text{for every } s > 0.$$

They state that a complete characterization of $\mathcal{X}$ is open. They prove that Lorentz spaces $\Lambda_r(\varphi)$ belong to $\mathcal{X}$, that the Marcinkiewicz spaces $M(t^\alpha)$ do not, and that a Δ_2 Orlicz space L_M belongs to $\mathcal{X}$ under a monotonicity assumption on $M'(\varepsilon x)/M'(x)$.

Result

This packet gives a necessary and sufficient condition for the Orlicz subcase under the same Δ_2 modular regime used by the source paper.

Let M be a strictly increasing Orlicz function and, for $0 < \varepsilon < 1$, put

$$Q_\varepsilon(u, v) = \frac{M(\varepsilon v) - M(\varepsilon u)}{M(v) - M(u)}, \quad 0 < u < v.$$

Theorem 1. *Let M be a strictly increasing Δ_2 Orlicz function on $[0, \infty)$. Then $L_M(0, \infty)$, equipped with the Luxemburg norm, belongs to $\mathcal{X}$ if and only if, for every $0 < \varepsilon < 1$ and every $0 < u < v < w$,*

$$Q_\varepsilon(u, v) \geq Q_\varepsilon(v, w). \tag{SC}$$

Thus the source paper’s differentiable condition is not merely a convenient sufficient condition; after replacing derivatives by secants it is the exact Orlicz-side dividing line.

Proof intuition

The proof is a measure-ratio version of the source paper’s Orlicz argument. For fixed ε , compare the Stieltjes measures

$$dM_\varepsilon(y) := d(M(\varepsilon y)), \quad dM(y).$$

Condition (SC) says that the ratio dM_ε/dM , interpreted through interval averages, decreases as the level interval moves to the right.

The NP crossing makes the distribution-function difference have one sign: the g -excess lives below the crossing level and the f -excess lives above it. If a high-scale modular comparison favors g , the decreasing ratio condition forces the lower-scale modular comparison to favor g as well. This is exactly the step needed to rule out a truncation failure.

Conversely, if (SC) fails, one can concentrate the lower-level excess and higher-level excess on two adjacent intervals where the ratio is in the wrong order. A two-step prefix plus a common low tail then gives an explicit NP pair with $\|f\|_{L_M} \geq \|g\|_{L_M}$ but with a failed prefix comparison.

The interval-ratio lemma

Let μ and μ_ε be the Stieltjes measures determined on intervals by

$$\mu((a, b]) = M(b) - M(a), \quad \mu_\varepsilon((a, b]) = M(\varepsilon b) - M(\varepsilon a).$$

Lemma 1. *Assume (SC). Fix $\tau > 0$ and $0 < \varepsilon < 1$. If A and B are nonnegative measurable functions with A supported in $(0, \tau)$ and B supported in (τ, ∞) , and if all displayed integrals are finite, then*

$$\int A d\mu > \int B d\mu \implies \int A d\mu_\varepsilon > \int B d\mu_\varepsilon.$$

Proof. Condition (SC) implies the following separated-interval form: whenever intervals I and J satisfy $\sup I \leq \inf J$ and $\mu(I), \mu(J) > 0$,

$$\frac{\mu_\varepsilon(I)}{\mu(I)} \geq \frac{\mu_\varepsilon(J)}{\mu(J)}. \quad (1)$$

Indeed, the ratio over a union of adjacent intervals is a weighted average of the ratios over the pieces, so the three-point condition propagates to arbitrary intervals by subdivision.

By finite simple-function approximation, (1) extends from intervals to measurable sets on opposite sides of τ , and then by the layer-cake formula to nonnegative functions. Thus

$$\frac{\int A d\mu_\varepsilon}{\int A d\mu} \geq \frac{\int B d\mu_\varepsilon}{\int B d\mu},$$

with the evident convention when one denominator is zero. Since M is strictly increasing, the relevant ratios are positive. The strict inequality $\int A d\mu > \int B d\mu$ therefore gives the claim. $\square$

Sufficiency

We prove that (SC) implies $L_M \in \mathcal{X}$. Write

$$\rho_M(h) = \int_0^\infty M(|h(t)|) dt$$

for the Orlicz modular. Under Δ_2 , the standard Luxemburg modular normalization gives $\|h\|_{L_M} = 1$ iff $\rho_M(h) = 1$ for nonzero h in L_M , and modular convergence is compatible with monotone truncations. As in the source paper's proof of its Orlicz theorem, it is therefore enough to argue first for continuous decreasing rearrangements with finite support and then pass to limits.

Let (f, g) satisfy the NP condition and $\|f\|_{L_M} \geq \|g\|_{L_M}$. Scale so that $\|f\|_{L_M} = 1$. Then $\rho_M(f) = 1$ and $\rho_M(g) \leq 1$.

Suppose, toward a contradiction, that the defining truncation inequality fails at some s_0 . It cannot fail for $s_0 \leq t_0$, so $s_0 > t_0$. Let $F = f^* \chi_{(0, s_0)}$ and $G = g^* \chi_{(0, s_0)}$. Since $\|F\|_{L_M} < \|G\|_{L_M} \leq 1$, choose λ with

$$\|F\|_{L_M} < \lambda < \|G\|_{L_M} \leq 1.$$

Set $a = 1/\lambda > 1$ and $\varepsilon = \lambda$. Then

$$\int_0^{s_0} M(aG(t)) dt > 1 \geq \int_0^{s_0} M(aF(t)) dt. \quad (2)$$

Let $\Phi = aF$ and $\Psi = aG$. The NP crossing implies that, for some level τ , the distribution-function difference has the sign pattern

$$\lambda_\Psi(y) - \lambda_\Phi(y) \geq 0 \quad (0 < y < \tau), \quad \lambda_\Phi(y) - \lambda_\Psi(y) \geq 0 \quad (y > \tau).$$

By the layer-cake/Stieltjes representation of the modular, inequality (2) says

$$\int_0^\tau (\lambda_\Psi - \lambda_\Phi) d\mu > \int_\tau^\infty (\lambda_\Phi - \lambda_\Psi) d\mu.$$

Applying Lemma 1 to $A = (\lambda_\Psi - \lambda_\Phi)_+$ below τ and $B = (\lambda_\Phi - \lambda_\Psi)_+$ above τ yields

$$\int_0^{s_0} M(G(t)) dt > \int_0^{s_0} M(F(t)) dt. \quad (3)$$

Since $s_0 > t_0$, the NP condition gives $g^*(t) \geq f^*(t)$ for $t > s_0$. Hence the tail modular of g is at least the tail modular of f . Combining this tail comparison with (3) gives $\rho_M(g) > \rho_M(f)$, contradicting $\rho_M(g) \leq \rho_M(f) = 1$. Therefore no truncation failure is possible, and $L_M \in \mathcal{X}$.

Necessity

It remains to show that failure of (SC) forces $L_M \notin \mathcal{X}$. Suppose that for some $0 < \varepsilon < 1$ and $0 < d < a < b$,

$$\frac{M(\varepsilon a) - M(\varepsilon d)}{M(a) - M(d)} < \frac{M(\varepsilon b) - M(\varepsilon a)}{M(b) - M(a)}. \quad (4)$$

Equivalently, there is $\theta > 0$ such that

$$\frac{M(b) - M(a)}{M(a) - M(d)} < \theta < \frac{M(\varepsilon b) - M(\varepsilon a)}{M(\varepsilon a) - M(\varepsilon d)}. \quad (5)$$

The left inequality gives

$$M(b) + \theta M(d) < (1 + \theta)M(a). \quad (6)$$

Choose $B > 0$ so that

$$B(M(b) + \theta M(d)) < 1 < B(1 + \theta)M(a), \quad (7)$$

and put $C = \theta B$ and $s_0 = B + C$.

On $(0, s_0)$ define decreasing step functions

$$U_0(t) = \begin{cases} b, & 0 < t < B, \\ d, & B < t < s_0, \end{cases} \quad V_0(t) = a \quad (0 < t < s_0).$$

Then (7) says

$$\int_0^{s_0} M(U_0(t)) dt < 1 < \int_0^{s_0} M(V_0(t)) dt. \quad (8)$$

The right inequality in (5) gives

$$BM(\varepsilon b) + CM(\varepsilon d) > (B + C)M(\varepsilon a). \quad (9)$$

Because $\varepsilon < 1$ and M is increasing, the left side of (9) is strictly smaller than $BM(b) + CM(d)$, hence is < 1 by (7). Therefore

$$T = \frac{1 - \int_0^{s_0} M(\varepsilon U_0(t)) dt}{M(\varepsilon d)}$$

is positive. Extend by a common low tail:

$$U(t) = \begin{cases} U_0(t), & 0 < t < s_0, \\ d, & s_0 < t < s_0 + T, \\ 0, & t > s_0 + T, \end{cases} \quad V(t) = \begin{cases} V_0(t), & 0 < t < s_0, \\ d, & s_0 < t < s_0 + T, \\ 0, & t > s_0 + T. \end{cases}$$

Finally set $f = \varepsilon U$ and $g = \varepsilon V$.

These functions are already decreasing rearrangements. With $t_0 = B$ they satisfy the source paper's NP condition: $f^* \geq g^*$ on $(0, B)$ and $f^* \leq g^*$ on (B, ∞) .

At scale 1, the choice of T gives

$$\int_0^\infty M(f(t)) dt = 1,$$

while (9) gives

$$\int_0^\infty M(g(t)) dt < 1.$$

Thus $\|f\|_{L_M} = 1 > \|g\|_{L_M}$. On the prefix $(0, s_0)$, (8) says

$$\int_0^{s_0} M(f(t)/\varepsilon) dt < 1 < \int_0^{s_0} M(g(t)/\varepsilon) dt,$$

so

$$\|f^* \chi_{(0, s_0)}\|_{L_M} < \varepsilon < \|g^* \chi_{(0, s_0)}\|_{L_M}.$$

This violates the defining implication for $\mathcal{X}$. Hence $L_M \notin \mathcal{X}$.

The sufficiency and necessity parts prove Theorem 1.

Consequences

Corollary 1. *The source paper's differentiable condition implies (SC). In particular, Theorem 1 recovers its positive Orlicz theorem.*

Proof. If M is differentiable and $x \mapsto M'(\varepsilon x)/M'(x)$ is nonincreasing, then

$$Q_\varepsilon(u, v) = \frac{\int_u^v \varepsilon M'(\varepsilon x) dx}{\int_u^v M'(x) dx} = \frac{\int_u^v \varepsilon M'(\varepsilon x)/M'(x) dM(x)}{\int_u^v dM(x)}.$$

Thus $Q_\varepsilon(u, v)$ is the dM -average over (u, v) of a nonincreasing function. Such averages decrease as the interval moves to the right, giving (SC). $\square$

Corollary 2. Let $0 < p < q$ and define

$$N_{p,q}(u) = \begin{cases} u^q, & 0 \leq u \leq 1, \\ u^p, & u \geq 1, \end{cases} \quad M_{p,q}(u) = \int_0^u N_{p,q}(v) dv.$$

Then $M_{p,q}$ is a Δ_2 Orlicz function and $L_{M_{p,q}}(0, \infty) \notin \mathcal{X}$.

Proof. The function $N_{p,q}$ is increasing and continuous, so $M_{p,q}$ is an Orlicz function, and its two polynomial growth regimes imply Δ_2 . Fix $0 < \varepsilon < 1$ and choose $0 < d < a < 1$. Then

$$\frac{M_{p,q}(\varepsilon a) - M_{p,q}(\varepsilon d)}{M_{p,q}(a) - M_{p,q}(d)} = \varepsilon^{q+1}.$$

As $b \rightarrow \infty$,

$$\frac{M_{p,q}(\varepsilon b) - M_{p,q}(\varepsilon a)}{M_{p,q}(b) - M_{p,q}(a)} \rightarrow \varepsilon^{p+1}.$$

Since $0 < p < q$ and $0 < \varepsilon < 1$, $\varepsilon^{q+1} < \varepsilon^{p+1}$. Hence (SC) fails for sufficiently large b , and the characterization gives $L_{M_{p,q}} \notin \mathcal{X}$. $\square$

Concrete sanity check

The script `code/verify_concrete.py` checks the explicit decreasing power instance $p = 1$, $q = 2$, $\varepsilon = 1/2$, $d = 1/4$, $a = 1/2$, $b = 20$. It prints

$$0.125 = \frac{M(\varepsilon a) - M(\varepsilon d)}{M(a) - M(d)} < \frac{M(\varepsilon b) - M(\varepsilon a)}{M(b) - M(a)} \approx 0.2494,$$

then constructs admissible B, C, T and verifies the prefix and global modular inequalities used in the necessity construction.

Limitations and novelty check

This packet does not solve the source paper's complete problem for all rearrangement invariant spaces. It gives a full characterization of the Δ_2 Orlicz subcase and turns the source paper's differentiable sufficient condition into an exact secant-ratio criterion.

On 2026-07-03, web searches for exact and close phrases around *Majorization revisited*, $\mathcal{X}$, Orlicz spaces, Nazarov–Podkorytov, and the ratio $M'(\varepsilon x)/M'(x)$ found the source arXiv page but no later answer or matching Orlicz characterization. Novelty confidence is moderate, pending expert literature review.

Human-review recommendation

The main points to check are Lemma 1, the use of the standard Δ_2 Luxemburg modular normalization, and the reduction from continuous decreasing rearrangements to the general Orlicz-space case. The necessity direction is explicit and finite-dimensional.

References

- [1] S. V. Astashkin, K. V. Lykov, and M. Milman, *Majorization revisited: Comparison of norms in interpolation scales*, arXiv:2107.11854, 2021.